BOARD OF INTERMEDIATE AND SECONDARY EDUCATION, SARGODHA

SECONDARY SCHOOL (9TH CLASS) EXAMINATION, 2022

PHYSICS

Past Paper 2022 • Paper Code: 1471 / 1472

Retyped faithfully from the original board paper — same questions, no changes.

GROUP – I

Q. Paper : (Objective Type)

سوالیہ پرچہ: (معروضی طرز)

QQ-1. 0922 (جماعت نہم) — سیکنڈری پارٹ I — Session 2018-20 to 2021-23 — فزکس (معروضی) — گروپ پہلا

Q. Time: 15 minutes — Total Marks: 12 — PAPER CODE 1471

Note:- You have four choices for each objective type question as A, B, C and D. The choice, which you think, is correct; fill that circle in front of that question number. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero mark in that question. Write PAPER CODE, which is printed on this question paper, on the both sides of the Answer Sheet and fill bubbles accordingly, otherwise the student will be responsible for the situation. Use of ink remover or white correcting fluid is not allowed

Q. Q-1

Q1. Amount of a substance in terms of number is measured in

(A) (B)

(C) (D)

Q2. Which of the following is a vector quantity?

(A) (B)

(C) (D)

Q3. Inertia depends upon

(A) (B)

(C) (D)

Q4. Weight is a

(A) (B)

(C) (D)

Q5. Two equal but unlike parallel forces having different line of action produce.

(A) (B)

(C) (D)

Q6. In SI-unit, the value of G is

(A) (B)

(C) (D)

Q7. Earth's gravitational force of attraction vanishes at

(A) (B)

(C) (D)

Q8. Efficiency of an ideal system is

(A) (B)

(C) (D)

Q9. The density of a substance can be found with the help of

(A) (B)

(C) (D)

Q10. Which of the following affects evaporation?

(A) (B)

(C) (D)

Q11. Land and Sea breezes are also the examples of

(A) (B)

(C) (D)

Q12. SI-Unit of specific heat is

(A) (B)

(C) (D)

Paper : (Essay Type)

پرچہ: (انشائیہ طرز)

QQ-2. 0922 (**جماعت نہم**) — **سیکنڈری پارٹ** I — Session 2018-20 to 2021-23 — **فزکس** (**انشائیہ**) — **گروپ پہلا** — **حصہ اول**

Q. (Marks 5×2=10) Time: 1:45 hours — Total Marks: 48 — Part I

سوال نمبر 2- درج ذیل میں سے کوئی سے پانچ اجزاء کے مختصر جوابات تحریر کریں۔ 5x2=10

(i) Estimate 20 years age in seconds. (ii) What is meant by significant figures? And how many significant figures are there in 0.027? (iii) Define base quantities and base units. (iv) Differentiate between distance and displacement. (v) A sprinter completes its 100 m race in 12 seconds. Find its average speed. (vi) What will be the effect on the value of centripetal force when the velocity of body becomes double? (Give reason). (vii) Define momentum and write its formula. (viii) When a gun is fired, it recoils why?

Q. (Marks 5×2=10) Q-3

سوال نمبر 3- درج ذیل میں سے کوئی سے پانچ اجزاء کے مختصر جوابات تحریر کریں۔ 5x2=10

(i) State second condition of equilibrium. Write its formula. (ii) Distinguish between like and unlike parallel forces. (iii) Why the value of "g" is different on the surface of the earth. (iv) Define geostationary orbit. (v) Write down the value of "G" and mass of earth with SI units. (vi) Define and write the Einstein's mass energy equation. (vii) What is meant by elastic potential energy? (viii) Define magma.

Q. (Marks 5×2=10) Q-4

سوال نمبر 4- درج ذیل میں سے کوئی سے پانچ اجزاء کے مختصر جوابات تحریر کریں۔ 5x2=10

(i) Define Pressure. Write its SI unit. (ii) State Archemedes Principle. (iii) Define elasticity. (iv) Change 300 K on Kelvin scale into Celsius scale of temperature. (v) Define Latent heat of vaporization. (vi) Write down any two factors on which rate of heat flow depends. (vii) What causes a glider to remain in air? (viii) Define conduction and convection modes of heat transfer.

Q. (Marks 9 × 2 = 18) Part II — Note: Attempt any Two Questions.

نوٹ: کوئی سے دو سوالات کے جوابات تحریر کریں۔

Q5. 5.a. Derive Third Equation of motion with the help of speed-time graph. 1+3

b. A body of mass 5 Kg is moving with a velocity of 10 ms^{-1}. Find the force required to stop it in 2 seconds. 5

Q6. 6.a. Determine a force from its perpendicular Components. 4

b. A motor boat moves at a steady speed of 4ms^{-1}. Water resistance acting on it is 4000 N. Calculate the power of its engine? 5

Q7. 7.a. Define Thermal conductivity of a substance and Prove that $K = Q/t \times L/(A(T_1-T_2))$ 1+3

b. An electric heater supplies heat at the rate of 1000 Joules per second. How much time is required to raise the temperature of 200 g of water from 20 °C to 90 °C?

> Note: Board typos kept verbatim: 'Archemedes', 'Estimate 20 years age', 'An electric heater'.

GROUP – II

Q. Paper : (Objective Type)

سوالیہ پرچہ: (معروضی طرز)

QQ-1. 0922 (**جماعت نہم**) — **سیکنڈری پارٹ** I — Session 2018-20 to 2021-23 — **فزکس** (**معروضی**) — **گروپ دوسرا**

Q. Time: 15 minutes — Total Marks: 12 — PAPER CODE 1472

Note:- You have four choices for each objective type question as A, B, C and D. The choice, which you think, is correct; fill that circle in front of that question number. Use marker or pen to fill the circles. Cutting or filling two or more circles will result in zero mark in that question. Write PAPER CODE, which is printed on this question paper, on the both sides of the Answer Sheet and fill bubbles accordingly, otherwise the student will be responsible for the situation. Use of ink remover or white correcting fluid is not allowed

Q. Q-1

Q1. Least count of screw gauge is

(A) (B)

(C) (D)

Q2. Which of the following unit is not a derived unit?

(A) (B)

(C) (D)

Q3. The speed of Falcon to fly is

(A) (B)

(C) (D)

Q4. Which of the following is a vector quantity?

(A) (B)

(C) (D)

Q5. The formula to find the centripetal force is

(A) (B)

(C) (D)

Q6. A force of 10 N is making an angle of 30° with horizontal. Its horizontal component will be

(A) (B)

(C) (D)

Q7. The value of 'g' on moon surface is 1.6 ms^{-2}. What will be the weight of a 100 kg body on the surface of the moon?

(A) (B)

(C) (D)

Q8. The numbers of total satellites in Global Positioning System are

(A) (B)

(C) (D)

Q9. The work done will be zero when the angle between the force and the distance is

(A) (B)

(C) (D)

Q10. Which of the substance is the lightest one?

(A) (B)

(C) (D)

Q11. Water freezes at

(A) (B)

(C) (D)

Q12. The faces of Leslie's cube are

(A) (B)

(C) (D)

Paper : (Essay Type)

پرچہ: (انشائیہ طرز)

QQ-2. 0922 (**جماعت نہم**) — **سیکنڈری پارٹ** I — Session 2018-20 to 2021-23 — **فزکس** (**انشائیہ**) — **گروپ دوسرا** — **حصہ اول**

Q. (Marks 5×2=10) Time: 1:45 hours — Total Marks: 48 — Part I

سوال نمبر 2- درج ذیل میں سے کوئی سے پانچ اجزاء کے مختصر جوابات تحریر کریں۔ 5x2=10

(i) A screw gauge has 50 divisions on its circular scale. The pitch of screw gauge is 0.5 mm. What is its least count? (ii) Define plasma physics and geophysics. (iii) Write down two rules

to identify significant figures. (iv) Differentiate between velocity and acceleration. (v) Define momentum and force. (vi) Write two methods to reduce friction. (vii) Why rolling friction is less than sliding friction? (viii) Convert 1 kmh^{-1} to ms^{-1}.

Q. (Marks 5×2=10) Q-3

سوال نمبر 3- درج ذیل میں سے کوئی سے پانچ اجزاء کے مختصر جوابات تحریر کریں۔ 5x2=10

(i) Define Axis of rotation. (ii) Why the height of vehicles is kept as low as possible? (iii) Differentiate between Torque and Couple. (iv) On what factors the orbital speed of a satellite depends? (v) What is meant by the force of gravitation? (vi) Law of gravitation is important to us, give reason. (vii) Define power and write its formula. (viii) Which form of energy is most preferred and why?

Q. (Marks 5×2=10) Q-4

سوال نمبر 4- درج ذیل میں سے کوئی سے پانچ اجزاء کے مختصر جوابات تحریر کریں۔ 5x2=10

(i) Write any two features of Kinetic molecular model of matter. (ii) Define Pressure and give its formula. (iii) What is meant by internal energy? (iv) What is Greenhouse effect? (v) Give any two properties of thermometric liquid. (vi) How does heating affect the motion of molecules of a gas? (vii) Write two uses of Convection currents. (viii) Why are metals good conductors of heat?

Q. (Marks 9 × 2 = 18) Part II — Note: Attempt any Two Questions.

نوٹ: کوئی سے دو سوالات کے جوابات تحریر کریں۔

Q5. 5.a. Derive third equation of motion by graphical method. 1+3

b. A body has weight 20 N. How much force is required to move it vertically upward with an acceleration of 2 ms^{-2}? 5

Q6. 6.a. Determine the mass of the Earth. 3+1

b. The steering of a car has a radius 16 cm. Find the torque produced by a couple of 50 N. 5

Q7. 7.a. Explain the working of a hydranlic system as a force multiplier. 4

b. A container has 2.5 litres of water at 20 °C. How much heat is required to boil the water. 5

> Note: Board typos kept verbatim: 'scle', 'followings', 'parte', 'hydranlic'.